FTML practical session 14

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Table des matières

1 Lower bound on the performance of gradient-based algorithms 1

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Exercice 1 :
Comppositions
Exercice 2 :
We note that $2t+1 \leqslant d$.

$$\nabla f(x) = \frac{L}{4}\begin{pmatrix} 2x_1 - x_2 - 1 \\ x_2 - x_1 + x_2 - x_3 \\ \dots \\ x_i - x_{i-1} + x_i - x_{i+1} \\ \dots \\ x_{2t-1} - x_{2t} \\ x_{2t} - x_{2t+1} \\ -(x_{2t} - x_{2t+1}) + x_{2t+1} \\ 0 \\ \dots \\ 0 \end{pmatrix} = \frac{L}{4}\begin{pmatrix} 2x_1 - x_2 - 1 \\ 2x_2 - x_3 - x_1 \\ \dots \\ 2x_i - x_{i+1} - x_{i-1} \\ \dots \\ 2x_{2t-1} - x_{2t} - x_{2t-2} \\ x_{2t} - x_{2t+1} \\ 2x_{2t+1} - x_{2t} \\ 0 \\ \dots \\ 0 \end{pmatrix} \in \mathbb{R}^d \quad (1)$$

Exercice 3 :
The result is true for $l = 0$. At iteration 0, all the components of the gradient are 0, except the first one, with a value of $-\frac{L}{4}$.
Hence, $x_j^1 = 0$ (iteration $l = 1$) for j $\geqslant 2$.
All the components of the gradient are 0 for $j \geqslant 3$.
Exercice 4 :

$$\begin{aligned} f(x^t) &= \frac{L}{4}\Big(\frac{1}{2}(x_1^t)^2 + \frac{1}{2}\sum_{i=1}^{2t}(x_i^t - x_{i+1}^t)^2 + \frac{1}{2}(x_{2t+1}^t)^2 - x_1^t\Big) && (2) \\ &= \frac{L}{4}\Big(\frac{1}{2}(x_1^t)^2 + \frac{1}{2}\sum_{i=1}^{t-1}(x_i^t - x_{i+1}^t)^2 + \frac{1}{2}(x_t^t)^2 - x_1^t\Big) && (3) \\ &= g(x^t) && (4) \\ & && (5) \end{aligned}$$

Exercice 5 :

$$
\begin{aligned}
\|x^*\|^2 &= \sum_{i=1}^{d} (x_i^*)^2 && (6)\\
&= \sum_{i=1}^{2t+1} (x_i^*)^2 && (7)\\
&= \sum_{i=1}^{2t+1} (1 - \frac{i}{2t+2})^2 && (8)\\
&= \sum_{i=1}^{2t+1} (\frac{2t+2-i}{2t+2})^2 && (9)\\
&= \sum_{j=1}^{2t+1} (\frac{j}{2t+2})^2 && (10)\\
&= (\frac{1}{2t+2})^2 \sum_{j=1}^{2t+1} j^2 && (11)\\
&= (\frac{1}{2t+2})^2 (2t+1)(2t+2)(4t+3)/6 && (12)\\
&= \frac{1}{2t+2}(2t+1)(4t+3)/6 && (13)\\
&\leqslant (4t+3)/6 && (14)\\
&\leqslant (4t+4)/6 && (15)\\
&= (2t+2)/3 && (16)\\
& && (17)
\end{aligned}
$$